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NEP 2020 FOLLOWS NCF-FS

A Textbook of
Computer Science with Lab Activities

Mastering CODING

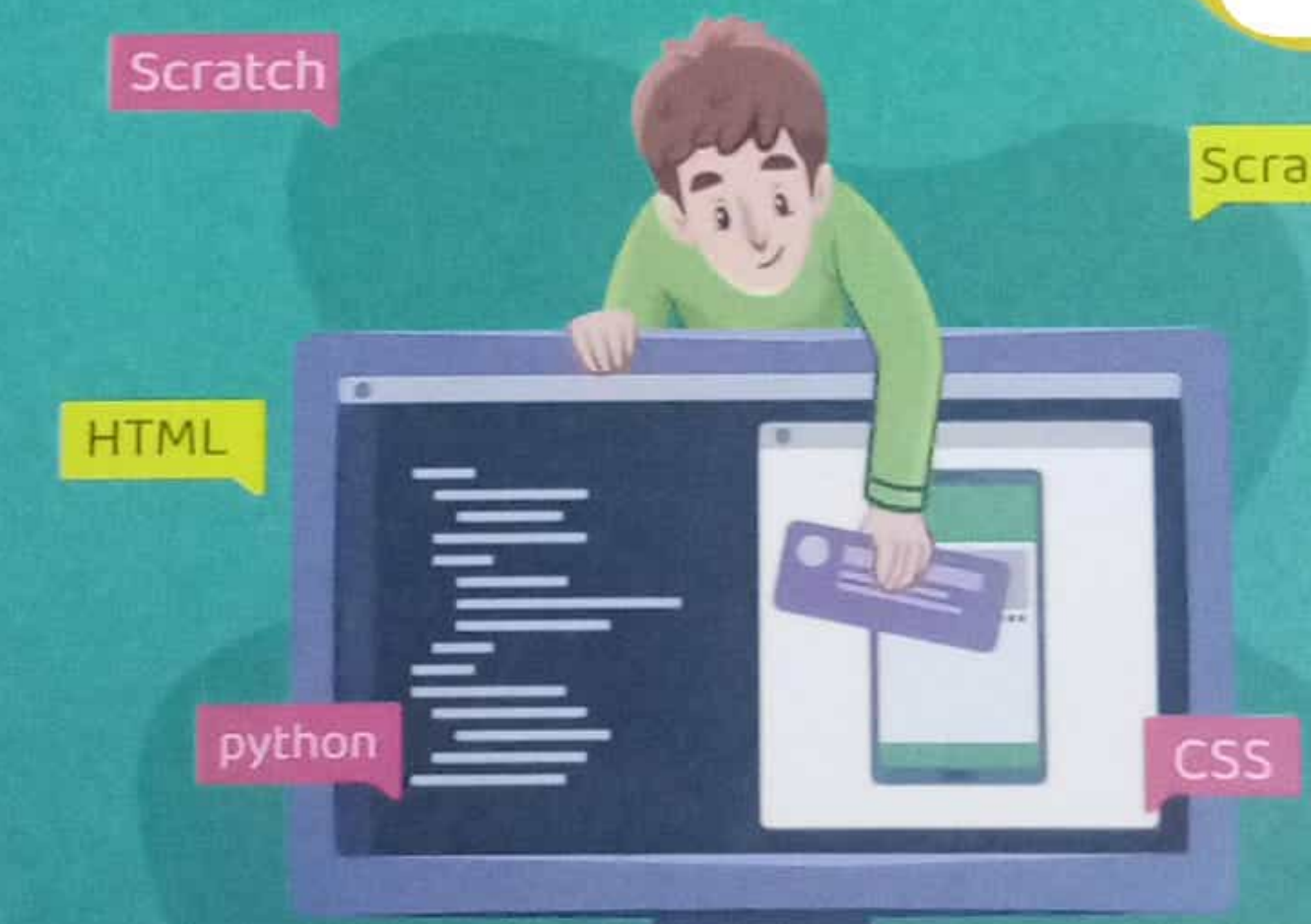
AI AND ROBOTICS

Windows 10
& MS Office 2016

FOUNDATIONAL STAGE
1

Scratch

ScratchJr



HTML

python

CSS

Mobile Application | Video Learning | AI- Powered Worksheets | Test Paper Generator

Scan Code for Hybrid Learning
TOKEN
76831

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PREFACE

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Mastering Coding—AI and Robotics is a book series designed for Grades I to VIII, offering a hands-on, interactive approach to computer science. The series emphasizes practical skills alongside theoretical knowledge, ensuring that learning remains engaging and enjoyable for students.

Key Features

1. **Dynamic Curriculum:** Regularly updated content reflects the latest trends in technology.
2. **Interactive Learning:** Hands-on exercises encourage exploration and experimentation.
3. **Holistic Approach:** Covers a wide range of topics, from digital literacy and internet safety to coding and artificial intelligence, all presented in an engaging manner.
4. **Student-Centric Tone:** Clear and friendly language caters to learners of all levels.
5. **Real-Life Applications:** Encourages students to apply their knowledge in practical scenarios.

We hope this book inspires students to explore the world of computers with curiosity and enthusiasm. Feedback and suggestions for improvement are always welcome.

—Publishers

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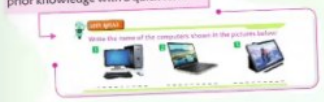

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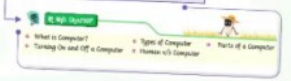

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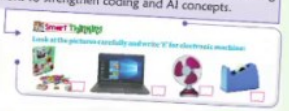

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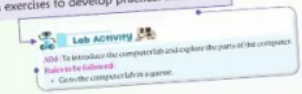

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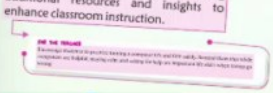

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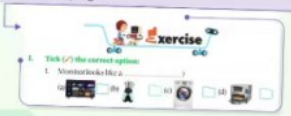

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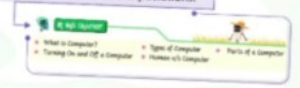

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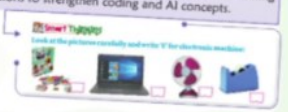

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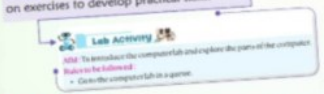

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CONTENTS

1.	More about a Computer	...	5
2.	Applications of Computers	...	13
3.	Keyboard and Mouse	...	20
4.	Parts of a Computer	...	32
5.	IPO Cycle	...	39
6.	Fun with Tux Paint	...	44
7.	Sequencing and Decoding	...	55
8.	Instructions and Direction Sense	...	63
9.	AI and Robotics	...	69
•	Periodic Test Term-1	...	75
•	Half Yearly Test Paper	...	76
•	Periodic Test Term-2	...	78
•	Annual Test Paper	...	79

CHAPTER 1

More about a Computer

LET'S RECALL

Write the name of the computers shown in the pictures below:

1

2

3

IN THIS CHAPTER

- What is Computer?
- Types of Computer
- Parts of a Computer
- Turning On and Off a Computer
- Human v/s Computer

WHAT IS COMPUTER?

A computer is an electronic machine that helps us perform many tasks quickly and without mistakes. We can listen to music, watch movies, type important documents, play games, draw, and solve sums on a computer. It has parts like a monitor, keyboard, mouse, and CPU. We can find computers in schools, banks, hospitals, and airports. They are very useful and are an important part of our lives.

TYPES OF COMPUTER

There are different types of computers that come in various shapes and sizes depending on their use, like personal computers, mobile computers, and wearable computers.

1. Personal Computers

These are commonly used at homes, offices, schools, and banks. Let's look at the different types of personal computers:

Desktop : These computers are placed on a desk, so they are called desktops.

Laptop : This is a portable computer that can be used on your lap, table, or anywhere you are comfortable. It is also good for use while travelling.

2. Mobile Computers

These are small-sized computers, some of which can also make calls, like smartphones.

Tablet: A portable mobile computer with a large touch screen.

Smartphone: A mobile phone with an integrated computer.

3. Wearable Computers

These are computers you can wear, like smartglasses or smartwatches. They are used for specific purposes and often have some limitations.

PARTS OF A COMPUTER

A computer consists of four main parts :

Keyboard

Mouse

Monitor

CPU

All these parts together make a computer. They are connected through wires to work together.

Monitor

Monitor is the screen that shows what you are doing on the computer. It is also called Visual Display Unit (VDU).

CPU

CPU is the brain of the computer. It processes information and controls everything. Its full form is Central Processing Unit. It controls all the other parts of a computer.

INFO Byte

The first computer ever made was called the ENIAC. It was so big that it filled an entire room.

Keyboard

Keyboard is a part of the computer that has many keys on it. These keys are used to write on the computer by pressing them. It is called typing. Whatever you type on the keyboard can be seen on the monitor.

Mouse

A mouse is a pointing device usually connected to a computer with a cable. It helps us point at things on the computer screen. We use the mouse to click, drag, and interact with the computer. It works best on a mouse pad because it moves smoothly on a flat surface.

Other devices connected to a computer are :

Printer

A printer is a machine that takes text and pictures from a computer and prints them onto paper. This printed paper is called a printout.

Speaker

Speakers are devices that play sound from a computer or other electronic devices, letting you hear music, voices, and other audio.

UPS

A UPS, or Uninterruptible Power Supply, is a device that provides backup power to a computer when the electricity goes out. It helps keep your computer running for a short time so you can save your work and shut down safely.

TURNING ON A COMPUTER

You have to follow certain steps to turn on a computer:

Step 1 : Plug in the computer into a power source and switch ON the power supply.

Step 2 : Switch on the UPS.

Step 3 : Locate and press the power button on the computer's CPU.

Step 4 : The computer will start up and display the start-up screen. Wait for it to load completely.

Step 5 : If your computer has a login screen, press any key on the keyboard. Enter your username and password and press enter key.

Step 6 : The home screen of your desktop will appear with small pictures on it. They are called icons.

SHUTTING DOWN A COMPUTER

Steps to shut down a computer are as follows :

Step 1 : Make sure to save any documents or files you are working on.

Step 2 : Close all open programs and windows.

Step 3 : Click the Start button in the bottom-left corner of the screen.

Step 4 : Choose the Shut Down option from the menu.

Step 5 : Wait for the computer to finish shutting down completely before turning off the power or unplugging it.

HUMAN VS COMPUTER

The computer is a smart machine. It can do a lot of things for us in a better way. Let's read the following:

Human	Computer
• Humans cannot work as fast as computer.	• Computers work at very fast speed.
• Humans need food, air and water.	• Computers need electricity to work.
• Humans can make mistakes.	• Computers never make mistakes.
• Humans get tired after working for sometime and get bored easily.	• Computers never get tired or bored.
• Humans are not multitaskers.	• Computers are multitaskers.
• Humans have limited memory and can forget after some time.	• Computers have a large memory and never forget.
• Humans have feelings.	• Computers do not have feelings.
• Humans can think and take decisions.	• Computers cannot take decisions on their own.

Let's Get Better

If your computer stops working while you are using it, what can you do? Write one way to stay calm and ask for help.

Activity Corner

Find the different types of computer.

D	B	C	D	L	E	F	G	H	I	J
D	K	L	T	A	B	L	E	T	M	N
E	O	P	Q	P	R	S	T	U	V	W
S	M	A	R	T	P	H	O	N	E	J
K	A	N	C	O	D	E	F	G	H	I
T	K	L	M	P	H	A	B	L	E	T
O	N	O	P	Q	R	S	T	A	B	C
P	D	Y	Z	X	Y	A	P	R	S	T

Exercise

I. Tick (✓) the correct option:

1. Monitor looks like a .....?

(a) (b) (c) (d)

2. What kind of device a computer is?

(a) Natural (b) Funny

(c) Electronic (d) Mechanical

3. Which of the following is a printer?

(a) (b)

(c) (d)

4. You can also carry it anywhere with you?

(a) (b)

(c) (d)

II. Fill in the blanks with the help of the given box.

1. A computer is a ..... device.
2. We use the ..... to point at different things on the screen.
3. ..... get tired after working sometime.
4. A ..... is very slim and has a larger touch screen than a smartphone.

tablet
human
mouse
math

III. Write 'T' for true and 'F' for false.

1. UPS stands for Uninterruptible Power Support.
2. Humans have limited memory and can forget after some time.
3. The speed of a computer is very slow, like a tortoise.
4. You can carry desktop computer anywhere with you.

IV. Answer the following questions.

1. What do you mean by computer?
2. What are the parts of a computer?
3. Write two difference between human and computer.

Smart Thinking

Look at the pictures carefully and write 'E' for electronic machine:

Lab Activity

AIM : To introduce the computer lab and explore the parts of the computer.

Rules to be followed :

- Go to the computer lab in a queue.
- Listen to the teacher and follow the directions.
- Carefully use the computer and equipments.

Procedure :

- Identify the parts of a computer in the lab.
- Write the name of equipments in the given spaces :

FOR THE TEACHER

Encourage students to practice turning a computer ON and OFF safely. Remind them that while computers are helpful, staying calm and asking for help are important life skills when things go wrong.

2 CHAPTER

Applications of Computers

Let's Recall

Tick (✓) the places where you think computer is used.

In this Chapter

- Uses of Computer
- Advantages of Computer
- Applications of a Computer
- Disadvantages of Computer

USES OF COMPUTER

We use many machines with computers in them.

These machines work smartly because of the computers and make our lives easier.

Earlier, we studied that computers are used for various purposes. Let us recall them and study in some detail.

Smart Thinking

Look at the pictures carefully and write 'E' for electronic machine:

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These machines work smartly because of the computers and make our lives easier.

Earlier, we studied that computers are used for various purposes. Let us recall them and study in some detail.

1. Learning : Computers help us find a lot of information on the internet. We can use educational websites and online courses to learn new things. Computers also help with homework and research.
2. Entertainment : We can play video games, watch movies, and stream music on computers. They provide access to a variety of media and entertainment platforms. Computers also let us watch videos and live streams.
3. Communication : Computers allow us to send emails and instant messages to friends and family. We can use video calls to see and talk to people anywhere in the world.
4. Work : People use computers to create and edit documents. They help in managing and organising data.
5. Shopping : We can buy clothes, books, electronic items, and more online using computers. Shopping websites and apps make it easy to purchase items.
6. Art : We can use computers to draw, design, and create digital art. They help us edit photos and videos with special software.

APPLICATIONS OF A COMPUTER

Computers are very helpful in our day to day life. They are used at many places to make work easier and faster. Some places where computers are used regularly are as follows :

1. Computers are used in schools and colleges for teaching and learning. Teachers use computers to make timetables and report cards. They are used in libraries to keep records of books and to read digital books.
2. Computers are used at railway stations and airports for reservation and cancellation of tickets. Train and flight timings are also maintained with the help of computers.
3. Computers help doctors keep patient records, perform medical tests and detect diseases. They are also used to monitor patients' health and prepare medical reports.

4. We use computers to watch movies, play games, and listen to music at home.
5. Banks use computers to manage accounts and details of customers. We can withdraw money from ATMs with the help of computers.
6. Scientists use computers for experiments and analysing data. Weather forecasting is done with the help of computers. Launching and controlling the movement of satellites in space is also done with the help of computers.
7. Computers are used in publishing houses for designing and printing books, newspapers and magazines.

Activity Corner

In the grid given below, encircle the names of eight places where computers are used.

BANK
SCHOOL
AIRPORT
HOME
HOSPITAL
SHOP
OFFICE
HOTEL

A	S	F	A	Q	H	J	K	A	L
H	O	S	P	I	T	A	L	I	U
O	D	J	S	W	S	C	E	R	T
F	F	H	S	F	C	G	C	P	B
F	B	O	H	G	H	E	H	O	A
I	H	T	O	R	O	C	E	R	N
C	T	E	P	S	O	V	B	T	K
E	M	L	D	A	L	H	O	M	E

INFO Byte

The first computer application was a simple calculator in the 1940s! Today, computers can do everything from helping us explore space to creating video games.

ADVANTAGES OF COMPUTER

1. Speed : A computer is superfast and can process information very quickly.
2. Accuracy : Computers perform tasks without making mistakes. It always gives right answers.
3. Storage : Computers can store a large amount of data and information like pictures, stories and videos.
4. Multitasking : Computers can perform many different types of tasks. It can be used for playing games, learning, drawing, etc.
5. Hardworking : Computers can work continuously and reliably without getting tired and bored.
6. Automatic : Computers can work automatically without human help.
7. Connectivity : Computers connect us to the internet and allow for easy communication with others anywhere in the world.

DISADVANTAGES OF COMPUTER

1. Dependence : Relying too much on computers can make it hard for us to do things without them.
2. Health Issues : Using computers for a long time can cause eye strain, bad posture, and other health issues.
3. Privacy Risks : Keeping personal information on computers can make it easy for hackers to steal it.
4. Cost : Buying and maintaining computers can be expensive.
5. Job Loss : Automation and computers can cause some people to lose their jobs in certain fields.
6. Distractions : Easy access to games, social media, and other distractions can make it harder to get work done.
7. No Emotions : Since a computer is a machine, it doesn't have any feelings and it also cannot think or make decisions.

Let's get BETTER

Computers are helpful, but can sometimes distract us. How can you make sure to finish your homework before using the computer for fun?

I. Tick (✓) the correct option :

1. A computer stores a large amount of ..... .
   (a) money ☐ (b) information ☐
   (c) water ☐ (d) None of these ☐
2. Which of the following things cannot be done with the help of a computer?
   (a) Play games ☐ (b) Share feelings ☐
   (c) Solve sums ☐ (d) Shopping ☐
3. Where are computers used for making report cards?
   (a) Schools ☐ (b) Banks ☐
   (c) Shops ☐ (d) Hospitals ☐
4. Which of these is used to withdraw money?
   (a) ATMs ☐ (b) A.C. ☐
   (c) Washing machine ☐ (d) Laptop ☐

II. Fill in the blanks with the help of the given box.

1. We use a computer at home to play ..... .
2. In shops, computers are used for making ..... .
3. Computers are used in schools to make ..... .
4. Hospitals use computers to detect ..... .

diseases
games
bills
timetables

III. Answer the following questions.

1. Write any two uses of computer.
2. Write any two disadvantages of a computer.
3. Write the names of any two places where computers are used.
4. Write any one way in which computers are used in hospitals.

IV. What is the child doing in given pictures?

....

Smart Thinking

Write the first letter of each picture in the space provider and find the secret word. One has been done for you.

1.

H O M E Home

2.

3.

4.

Lab Activity

(a) Ask your teacher to show how he/she uses a computer to keep records of your class.

(b) Visit your computer lab to play some games on the computer with your classmates.

FOR THE TEACHER

Visit a place where computers are used, like banks, supermarkets etc., with your students.

CHAPTER 3

Keyboard and Mouse

Let's Recall

Write the names of the keys shown in the pictures below:

In this Chapter

- Keyboard
- Mouse

KEYBOARD

In the last class, you learned about some basic keys on a keyboard. These include the letter keys, number keys, spacebar, enter key, delete key, and arrow keys. The picture below shows how these keys are arranged on a typical keyboard.

The keys on the keyboard can be divided into several groups based on function :

- Typing keys
- Function keys (F1 - F12)
- Arrow keys
- Windows key
- Numeric keypad
- Enter key

INFO Byte

The 'QWERTY' layout on your keyboard was designed to prevent typewriter jams. Today, it helps us type faster and easier on computers.

Note : Typing keys include letter, punctuation, number and symbol keys.

Detailed functions of typing keys are as follows:

- The keys that show the letters A to Z are called alphabet keys.
- There are ten number keys (0 to 9) on the keyboard.
- You can also find number keys on the right side of the keyboard.
- The spacebar is the longest key on the keyboard. It is used to add space between two numbers or words.
- The enter key is used to move the cursor to the beginning of the next line. There are two enter keys on a keyboard.
- The backspace key is used to delete the character before the cursor or the selected text.
- While typing, a small blinking line appears on the screen. This is called the cursor.
- The arrow keys are used to move the cursor up, down, left, and right.

Let us learn about some more special keys of the keyboard.

Caps Lock Key

This key is used to type capital letters. It is located on the left side of the alphabet keys. When you press the Caps Lock key, a green light turns on. It indicates that the caps lock is active and now you can type in capital letters.

If you press it again, the green light will be turned off, and you can now type in small letters.

Combination Keys

Some keys on the keyboard perform special tasks when used with other keys. For example : 1. Shift Key 2. Control Key

Shift Key

There are two Shift keys on the keyboard. Both Shift keys do the same job, and you can use either one.

Key Combinations

Have you noticed the symbols on top of the number keys? To type these symbols, you need to use key combinations like this :

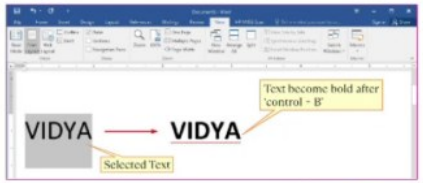
- Press the Shift key and the number key 5 together. It will type the symbol on the number key.

- Press the Shift key and the letter key A together. It prints the letter in capital.

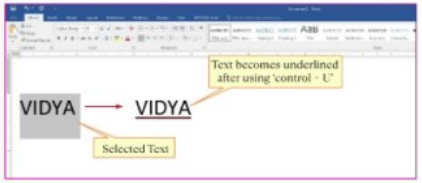
Note : Turn on the Caps Lock and then press shift key and letter key together. What did you notice? Discuss it with your teacher.

Control Key

There are two control keys on the keyboard. The Control key is labeled as Ctrl. You can use the Control key in a similar way to the Shift key. Using Ctrl key combinations provides shortcuts for many common tasks. For example : Ctrl + B and Ctrl + U



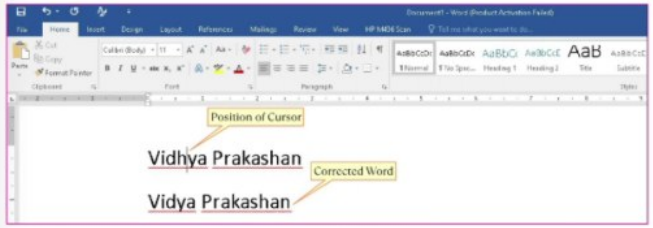
Makes the selected text bold.



Underlines the selected text.

Backspace

The Backspace key works like an eraser. If you type something wrong, you can delete it with the Backspace key. When you press the Backspace key, it erases the characters to the left of the cursor. For example :



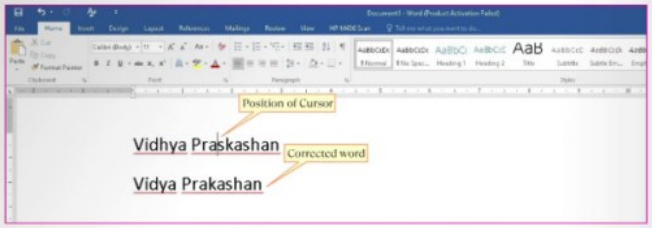
The word 'Vidya' is typed incorrectly. Let's correct it using the Backspace key :

- Place the cursor before the letter 'y'.
- Press the Backspace key.

This will delete the letter 'h' and correct the word to 'Vidya'.

Delete Key

The Delete key is used to delete characters to the right of the cursor. It is often abbreviated as the Del key. For example :



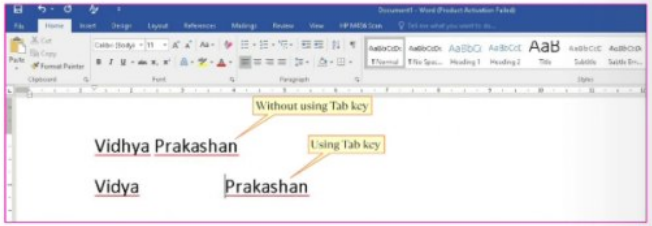
In the text above, the word 'Prakashan' is typed incorrectly. Let's correct it using the Delete key:

- Position the cursor after the letter 'a'.
- Press the Delete key.

This will delete the letter 's' and correct the word to 'Prakashan'.

Tab Key

The Tab key is used to add multiple spaces at once or move to a specific position in a document. When you press the Tab key, the cursor moves half an inch forward along with the text. For example :



Punctuation Keys

Punctuation marks are symbols used in sentences to help us understand them better. They include full stop, comma, colon, question mark, hyphen, etc. A simple paragraph might have several punctuation marks. You can find punctuation marks and symbols on a keyboard. If you look closely, you will see that some keys have two punctuation marks.

For example : The full stop (.) and greater than (>) symbols are on the same key. Let us learn how these punctuation keys are used while typing.

Punctuation Keys

Full Stop (.)

The full stop (.) is located on the keyboard near the middle to the right of the comma key. Find the key with the full stop on it. Press this key when you want to add a full stop to your text. It is commonly used to end sentences.

Comma

The comma (,) key is located to the left of the full stop key on the keyboard. Find the key with the comma on it. Press this key when you want to add a comma to your text. It is used to separate parts of a sentence or list.

Question Mark

The question mark (?) key is located to the right of the full stop key. It is on the same key as the forward slash (/) symbol. To use it :

- Hold down the Shift key.
- Press the key with the forward slash (/) to type a question mark.
- Use it to end questions in your text.

So we can notice that on keys with two symbols:

- The symbol on the lower part of the key can be typed by pressing the key normally.
- To type the symbol on the upper part of the key, press and hold the Shift key while pressing the same key.

Below is the table showing some more special symbols that we use while typing.

Symbol	Name	Symbol	Name
.	Full Stop	;	Semicolon
,	Comma	" "	Quotation Marks
!	Exclamation Mark	'	Apostrophe
?	Question Mark	( )	Parentheses
:	Colon	-	Hyphen

Activity Corner

In the given keyboard, the keys are marked with different colours.

Referring the colours, write the names of the keys in the given boxes :

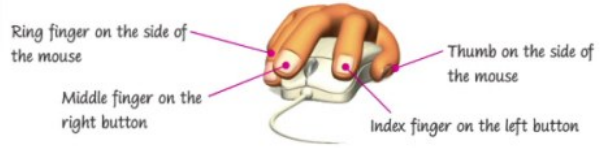
MOUSE

A mouse is a small device you use with your hand to move a pointer on the computer screen. It has buttons that you can click to open things and make the computer do different tasks. It helps you with activities like drawing and playing games.

Most mouse have three main parts :

- Left Button
- Right Button
- Scrolling Wheel

Holding the Mouse : Follow the methods shown in the figure to hold the mouse.



Left Click : This means pressing the left button on the mouse. It is like tapping to select or open things on the computer.

Right Click : This means pressing the right button on the mouse. It gives you special options, like copying or pasting.

Scrolling : This means using the wheel on the mouse to move up or down on pages.

Dragging : This involves holding down one of the buttons while moving the mouse to move or select something on the screen.

Let's Get BETTER

Using a keyboard and mouse can be fun, but it's important to take breaks and not spend too much time on the computer. What is one activity you can do away from the computer to keep yourself active and healthy?

I. Tick (✓) the correct option :

1. Which of the following characters will display with this combination?

(a) 2 (b) @ (c) @ 2 (d) 2@

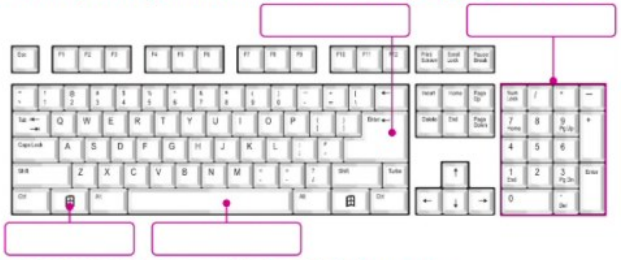
2. Which key is used to move the cursor?
   (a) Delete key (b) Arrow key
   (c) Backspace key (d) None of these
3. Which key is used to delete the characters on the right side of the cursor?
   (a) Basckspace (b) Delete
   (c) Spacebar (d) All of these
4. Which is the longest key on the keyboard?
   (a) Spacebar key (b) Enter key
   (c) Arrow key (d) None of these
5. How many Enter keys are present on the keyboard?
   (a) 3 (b) 2 (c) 0 (d) 1

6. How many number keys are present on the keyboard?
   (a) 20 (b) 18 (c) 21 (d) 10

II. Complete the following key combinations :

1. To type capital letter G : ..... + G
2. To make the text bold : Ctrl + ...
3. To underline the text : ..... + U

III. Identify the keys and write their names in the appropriate box :



IV. Match the following symbols with their names.

	Symbol	Name
1.	?	semicolon
2.	;	parentheses
3.	!	question Mark
4.	( )	comma
5.	,	exclamation mark

V. Write short notes on :

Delete

Ctrl

Tab

Shift

Smart Thinking

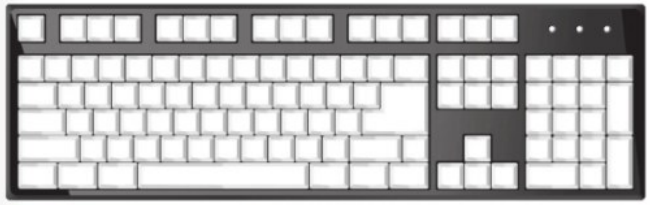
Colour the following keys with different colours in the given keyboard.

(a) Function keys (b) Number keys (c) Enter keys
(d) Longest key (e) Arrow keys

A blank computer keyboard is given to you. Now, perform the tasks as per the given instructions :

(a) Mark all the letters from A-Z in capitals as per the computer keyboard.

(b) Mark all the four types of arrow keys.
(c) Colour Enter key with blue.
(d) Colour Shift key with green.



Lab Activity

Write the following in Notepad/Wordpad.
(a) Your home address.
(b) Your father's and mother's name.
(c) The name of your school.

FOR THE TEACHER

Help the students in opening the Notepad/Wordpad program.

4 CHAPTER

Parts of a Computer

Let's Recall

Write the names of the devices given below :

In this Chapter

- Input Devices
- Output Devices
- Storage Devices

We are familiar with the different parts of a computer. On the basis of their functions, we can divide them into three categories.

- Input Devices
- Output Devices
- Storage Devices

INPUT DEVICES

Devices that let you give information or commands to a computer are called input devices. They help you communicate with the computer and enter data. Some of the input devices are discussed below :

Keyboard

We know keyboard is used to type letters and numbers. It helps us to give instructions to the computer.

Mouse

Mouse is an input device which is used to point, click, and scroll on the screen. It also helps us to select and open an application on the computer.

Microphone

A microphone is used to record sound on the computer.

Scanner

A scanner is used to copy pictures or documents into the computer.

Webcam

A webcam is a small camera that is connected to a computer. It lets you take pictures or record videos and is often used for video calls.

Activity Corner

Draw a picture of keyboard and colour the keys in groups according to their function.

OUTPUT DEVICES

Devices that show you the results from a computer are called output devices. They help you see, hear, or get information from the computer. Some of the output devices are discussed below:

Monitor

It displays pictures, videos, and text on a screen. There are different types of monitors.

LCD monitor

CRT monitor

Touch Screen monitor

Printer

It prints documents and pictures from the computer screen on to paper. This printed output is called a printout.

Speakers

We can listen to music on the computer through speakers. They let us hear the sound of songs, movies, and videos.

Headphones

Headphones are also used to listen to sound from the computer. When using speakers, people around you can hear the sound too. However, when you use headphones, only you can hear the sound. So the headphones let you listen to sound privately.

Activity Corner

Look at the following figure and make a list of the input and output devices.

Input Devices	Ouput Devices
1.	1.
2.	2.
3.	3.

STORAGE DEVICES

Devices that are used to save and store information are called storage devices. They keep your data safe so you can use it later. Some devices are fixed in the CPU, while others are removable. Some of the storage devices are discussed below.

Hard Disk

A hard disk is like a box that is fixed inside the CPU cabinet. When you save a file on the computer, it is stored on the hard disk. It holds a lot of data inside the computer.

INFO Byte

The first hard drive was the size of a refrigerator and could hold only 5 megabytes of data? Today, we can carry thousands of songs and games in our pockets on devices smaller than a smart phone!

USB Flash Drive

A USB flash drive is also called a pen drive. It helps you store data and carry it with you. It is easy to use and small enough to fit in your pocket. To use it, we plug it into the CPU cabinet.

CD/DVD

Both CDs and DVDs are round discs and are the same size, but a DVD stores more information than a CD. They both store less data than a pen drive.

Memory Card

Memory cards are small chips used in cameras and phones to store photos and videos.

These devices help you keep your files, photos, videos, and other important information safe and sound.

Let's Get Better

Computers have many parts that help us do different things. If you were to explain to a friend why it's important to take care of your computer and its parts, what would you say?

I. Fill in the blanks :

1. ..... devices allow us to enter data and information in the computer.
2. A ..... is used to record our sound in the computer.
3. A ..... is used to print the work done on the computer.
4. ..... are used to listen to the music by one person.
5. A ..... is easily carried in your pocket.

Input
Microphone
Headphones
Pen drive
Printer

II. Write 'T' for true and 'F' for false.

1. Mouse helps us to open an application on the computer.
2. A mouse and a keyboard are pointing devices.
3. A scanner is used to scan the pictures into the computer.
4. Monitor displays the work done on the computer.
5. We can connect on a video call with our friends using printer.

III. Who am I?

1. I have many buttons, called keys. .....
2. I help you record sound. .....
3. I help scan pictures. .....
4. I help you listen to music on the computer. .....
5. I help you store data and information. I am fixed inside the CPU cabinet. .....

IV. Unscramble the words.

1. EYKBOARD : .....
2. TERPRIN : .....
3. OTORNIM : .....
4. SPEASREK : .....
5. EDVPRENI : .....

V. Match the following :

1. Scanner
2. Monitor
3. Mouse
4. Webcam
5. CD/DVD

VI. Answer the following questions :

1. Give two examples of input and output devices.

2. What is the use of a webcam?
3. Which device can scan the images and documents and store in the computer?
4. Which part of a computer displays data and instructions?
5. Which type of device is a memory card?

Smart Thinking

Identify the given images and complete the crossword.

Lab Activity

Make a list of input devices, output devices and storage devices that you see in computer lab and at home.

	Input Devices	Output Devices	Storage Devices
Computer Lab			
Home			

FOR THE TEACHER

The various parts of a computer and their uses could be discussed in the class.

IPO Cycle

Let's Recall

Identify and tick (✓) the machines in the pictures below :

In this Chapter

- Working of Machines
- Working of a Computer

In the previous class, we have learnt that machines are tools that help us do work more easily. They make our lives easy. We use different machines every day, like bicycles to go places, washing machines to clean clothes, and fans to keep us cool. In this chapter, we will learn how machines work and follow the IPO cycle to help us in our daily lives.

WORKING OF MACHINES

A machine works on the simple process of Input Process Output cycle. They take something in, do some work, and give us a result. Look how this happens :

Washing Machine

INPUT	PROCESS	OUTPUT
Dirty clothes	Washing	Clean clothes

Juicer

INPUT	PROCESS	OUTPUT
Fruits	Juicer	Juice

In the above examples, we can clearly see that the machines took something in, did some work, and gave us a result.

Let's Get Better

If a machine stops working, what is one thing you can do to fix the problem instead of feeling upset?

WORKING OF A COMPUTER

We have also learnt that a computer is also a machine. It is a special kind of machine that helps us do many things, like play games, watch videos, write stories, and even solve math problems.

Just like other machines, a computer takes something in, works on it, and gives us a result. The process of IPO cycle is also followed by the computer.

Let us look at an example :

INPUT	PROCESS	OUTPUT
5, 2, -	5 - 2	3
Digits, Instruction	Subtraction	Result

Here, first we gave some instruction to the computer. Then computer took its time to follow the instruction and at last it gave us the result. Let us learn about it in detail.

Input

Input is the information or instructions you give to a machine or computer to make it work. For example, when you type on a keyboard, click a mouse, or speak into a microphone, you are giving input.

Process

Processing is what the machine or computer does with the input. After getting the input, the computer uses the Central Processing Unit (CPU) to take action. The CPU is the brain of the computer and does all the calculations.

Output

Output is the result you get from a machine or computer after it has processed the input. For example, when you see text on the screen, hear

INFO Byte

Computers can solve problems superfast! They can do millions of things in just one second.

music from the speakers, or get a printed page, those are all outputs. Output device helps us to get the output.

Exercise

I. Tick (✓) the correct option :

1. The working of a computer is based on the ..... cycle.
   (a) IPU ☐ (b) IPO ☐
   (c) IOP ☐ (d) None of these ☐
2. A computer performs action on the input with the help of ..... .
   (a) CPU ☐ (b) Printer ☐
   (c) Mouse ☐ (d) Scanner ☐
3. Which of these is an input device?
   (a) Keyboard ☐ (b) Monitor ☐
   (c) Printer ☐ (d) All of these ☐
4. The data or instructions we give to the computer is called ..... .
   (a) Input ☐ (b) Output ☐
   (c) Process ☐ (d) Method ☐
5. Which part of a computer works like a human brain?
   (a) CPU ☐ (b) Speakers ☐
   (c) Monitor ☐ (d) Printer ☐

II. Fill in the blanks with the help of the given box :

Mouse
Monitor
Keyboard
Processing
CPU

1. ..... is known as the brain of computer.
2. ..... shows us the result after processing.
3. ..... and ..... are the examples of input devices.
4. When a computer works on the data, it is called ..... .

III. Answer the following questions :

1. What is the full form of IPO?
2. Write the names of any two input and output devices.

Smart Thinking

Write input, process and output under the pictures given below :

1.

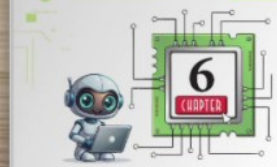
2.

Lab Activity

Look at the computer system in your lab. Find out which parts are used for input, process and output.

FOR THE TEACHER

Explain the concept of IPO cycle with the help of real-life examples.



Fun with Tux Paint

Let's Recall

On which of the following programs can we draw?

In this Chapter

- Introduction to Tux Paint
- Tools of Tux Paint
- Open, Save, Quit

INTRODUCTION TO TUX PAINT

Tux Paint is a fun and easy-to-use drawing program designed for young students like you. It lets kids create amazing pictures using a variety of tools and colours. When you open Tux Paint, you see a big, empty canvas ready for you to draw on, along with a cute penguin named Tux who helps you along the way.

How to Open Tux Paint?

Opening Tux Paint is easy and fun! Follow these simple steps to start drawing:

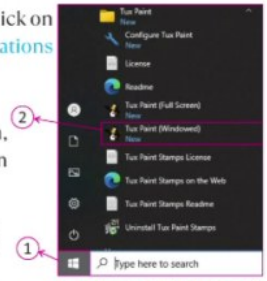
1. Find the Tux Paint Icon

- Look for the Tux Paint icon on your computer's desktop. It looks like a cute penguin.

- If you do not see it on the desktop, click on the Start button (Windows) or Applications (Mac) and search for Tux Paint.

2. Double-Click the Icon

- Once you find the Tux Paint icon, double-click on it using the left button of your mouse.
- If you are using a tablet or touch screen, you can tap on the icon twice quickly.



Select Tux Paint (Windowed)

Let's Get Better

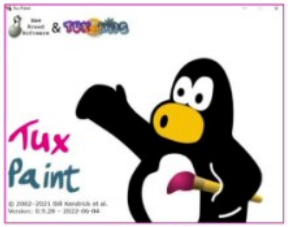
What is the difference between double-click and single-click? Explain the use of double-click and single-click while drawing.

3. Wait for Tux Paint to Open

After you double-click the icon, Tux Paint will start to open. You might see a small loading screen with the penguin, Tux.

4. Start Drawing

When Tux Paint opens, you will see a big blank canvas and lots of colourful tools. You are ready to start drawing and having fun!



Welcome Screen of Tux Paint

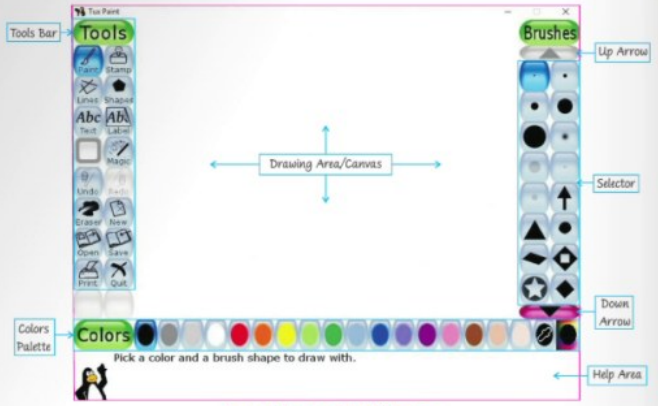
TOOLS OF TUX PAINT

1. Toolbar

- The toolbar is where you find all the drawing and editing tools.
- It is located on the left-hand side of your screen.
- You can use the toolbar to choose brushes, shapes, stamps and more.

2. Colour Palette

- The colour palette is where you pick the colours for your drawing.



Parts of the Tux Paint Window

- It has many different colours to choose from.
- You can find it at the bottom of the screen.

3. Selector

- The selector is used to choose different options for the tool you are using.
- It helps you change the size, style or type of the tool.
- You can see it on the right side of the toolbar.

4. Drawing Canvas

- The drawing canvas is the big blank space in the middle where you create your picture.
- This is your workspace for drawing, colouring and making art.

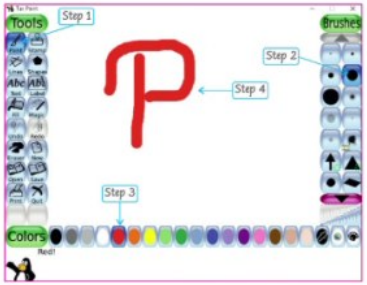
These parts help you use Tux Paint easily and make beautiful drawings.

Tux Paint Tools

Let us learn about the main tools in Tux Paint and how to use them :

Paint Tool

It draws freehand lines and shapes.



Step 1. Click on the brush icon in the toolbar.

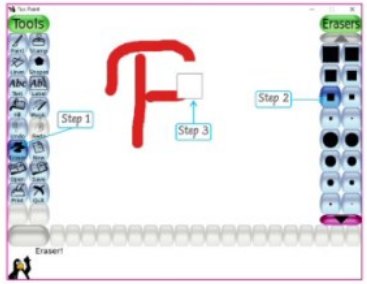
Step 2. Choose a brush style and size from the selector on the right.

Step 3. Pick a colour from the colour palette.

Step 4. Click and drag your mouse on the drawing canvas to draw.

Eraser Tool

This tool erases parts of your drawing.



Step 1. Click on the eraser icon in the toolbar.

Step 2. Choose an eraser size from the selector on the right.

Step 3. Click and drag your mouse on the drawing canvas to erase.

Text Tool

It adds text to your drawing.

Step 1. Click on the text icon in the toolbar.

Step 2. Choose a font from the selector on the right.

Step 3. Pick a colour from the colour palette.

Step 4. Click on the drawing canvas where you want the text to appear.

Step 5. Type your text using the keyboard.

Shapes Tool

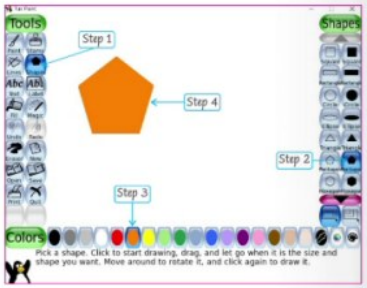
It draws shapes like circles, squares, and triangles.

Step 1. Click on the shapes icon in the toolbar.

Step 2. Choose a shape from the selector on the right.

Step 3. Pick a colour from the colour palette.

Step 4. Click and drag your mouse on the drawing canvas to draw the shape.



Stamp Tool

It adds pre-made images to your drawing.

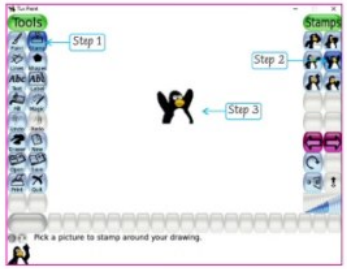
Step 1. Click on the stamp icon in the toolbar.

INFO Byte

Tux Paint is not just fun; it helps you become a great artist. You can use tools like stamp, brushes and shapes to make your own creative master piece.

Step 2. Choose a stamp from the selector on the right.

Step 3. Click on the drawing canvas to place the stamp.



Magic Tool

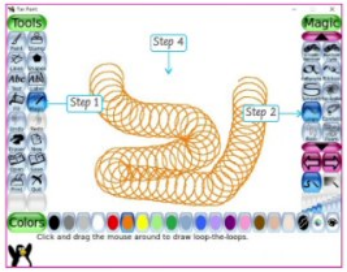
This tool adds special effects to your drawing.

Step 1. Click on the magic wand icon in the toolbar.

Step 2. Choose a magic effect from the selector on the right.

Step 3. Follow the instructions for the specific effect.

Step 4. Click and drag your mouse on the drawing canvas to apply the effect.



By using these tools, you can create amazing and colourful drawings in Tux Paint.

Save Tool

This tool saves your current drawing. You can save your drawing by just clicking on the save button in the Toolbar.

Let's Get BETTER

When you are using Tux Paint to create art, how can you make sure to share your work with friends or family? What steps can you take to save your pictures safely?

Open Tool

It opens a previously saved drawing.

Step 1. Look for the folder icon on the right side of the toolbar. Click on it to open a saved drawing.

Step 2. A screen with thumbnails of your saved drawings will appear. Click on the drawing you want to open.

Step 3. After selecting, click the Open button at the bottom of the screen. Your selected drawing will open on the drawing canvas.

Quit Tool

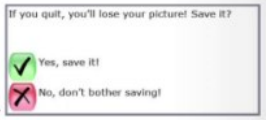
To close the Tux Paint, click on quit tool.

Step 1. Look for the red X or door icon on the right side of the toolbar.

Step 2. Click on it to quit Tux Paint.

Step 3. A message will appear asking if you really want to quit.

Step 4. Click Yes, save it! to close Tux Paint.



Step 5. If you have unsaved changes, Tux Paint will remind you to save your work before quitting.

These steps help you manage your drawings efficiently in Tux Paint, ensuring your work is saved, you can access previous drawings, and you can exit the program when finished.

Activity Corner

Find the following tools in the word maze given below :

B	E	S	T	A	M	P	N	I	Y
M	C	E	R	A	S	E	R	P	H
S	H	A	P	E	R	O	Q	H	C
A	H	N	S	A	V	E	I	E	V
G	T	O	P	E	N	T	Y	J	P
T	F	B	R	U	S	H	O	T	E

BRUSH
ERASER
SAVE
STAMP

Exercise

I. Tick (✓) the correct option :

1. The largest part of the Tux Paint window is ..... .
   (a) drawing canvas (b) toolbar
   (c) colours (d) None of these
2. ..... tool is used to type text on the drawing canvas.
   (a) Save (b) Brush
   (c) Paint (d) Abc Text
3. The Tux Paint window is divided in ..... main parts.
   (a) one (b) two
   (c) three (d) four
4. ..... tool is like a set of stickers.
   (a) Paint (b) Shapes
   (c) Stamp (d) None of these

Open Tool

It opens a previously saved drawing.

Step 1. Look for the folder icon on the right side of the toolbar. Click on it to open a saved drawing.

Step 2. A screen with thumbnails of your saved drawings will appear. Click on the drawing you want to open.

Step 3. After selecting, click the Open button at the bottom of the screen. Your selected drawing will open on the drawing canvas.

Quit Tool

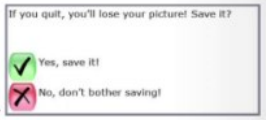
To close the Tux Paint, click on quit tool.

Step 1. Look for the red X or door icon on the right side of the toolbar.

Step 2. Click on it to quit Tux Paint.

Step 3. A message will appear asking if you really want to quit.

Step 4. Click Yes, save it! to close Tux Paint.



Step 5. If you have unsaved changes, Tux Paint will remind you to save your work before quitting.

These steps help you manage your drawings efficiently in Tux Paint, ensuring your work is saved, you can access previous drawings, and you can exit the program when finished.

Activity Corner

Find the following tools in the word maze given below :

B	E	S	T	A	M	P	N	I	Y
M	C	E	R	A	S	E	R	P	H
S	H	A	P	E	R	O	Q	H	C
A	H	N	S	A	V	E	I	E	V
G	T	O	P	E	N	T	Y	J	P
T	F	B	R	U	S	H	O	T	E

BRUSH
ERASER
SAVE
STAMP

Exercise

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1. The largest part of the Tux Paint window is ..... .
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   (a) one (b) two
   (c) three (d) four
4. ..... tool is like a set of stickers.
   (a) Paint (b) Shapes
   (c) Stamp (d) None of these

II. Fill in the blanks with the help of the given box :

Save
Paint
Stamp tool
Shapes
Sizes

1. ..... tool is used to draw free hand drawings.
2. ..... is used to paste pre-drawn images in the picture.
3. Click on ..... button, to save drawing.
4. Eraser tool is available in different ..... and ..... .

III. Unscramble the word :

1. O L O R U C .....
   The palette available at the bottom of the main screen.
2. C T O R S E L E .....
   It shows the different options when a tool is selected.
3. R A S E E R .....
   This tool is used to erase a part or whole of the drawing.

IV. Write 'T' for true and 'F' for false :

1. The selector section is located on the right side of the main screen.
2. The paint tool is used to add special effects to a picture.
3. The stamp tool is like a set of stickers.
4. The save button is used to save a picture in Tux Paint.

V. Who am I?

1. I am the button to save the current drawing. .....
2. I am the tool used to erase a drawing. .....

VI. Answer the following questions :

1. Write the steps to open Tux Paint.
2. What is Tux Paint?

3. What is the use of Paint tool in Tux Paint?
4. Which part of Tux Paint window is used for drawing?

Smart Thinking

Calculate and colour the picture using the colour code.

Colour Code

16 - Red	18 - Orange
20 - Brown	22 - Blue
24 - Yellow	26 - Green

Lab Activity

Draw the following pictures using Tux Paint.

FOR THE TEACHER

- Explain the steps to open Tux Paint.
- Explain how to change the size and colour of the brush while drawing.
- Explain how to search for a picture in the Stamp tool and paste it in the drawing.
- Explain how to make corrections in the drawing by using Eraser tool.

CHAPTER 7

Sequencing and Decoding

Let's Recall

1. Complete the following :

(a) Monday, ... , ... , Thursday, ... , ... , ...

(b) ..... , ...

2. Which of the following is the example of a loop?

In this Chapter

- Sequence
- Patterns and Loops
- Decision Making
- Decoding

SEQUENCE

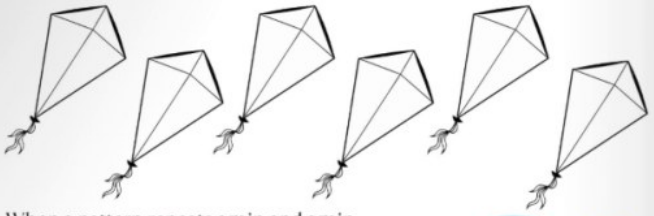
A sequence is a list of things in a specific and correct order. For example, a sequence can be a list of steps you need to follow to complete a task, like a recipe or instructions for building something.

January → February → March → April → May → June

PATTERNS AND LOOPS

A pattern is something that repeats in a regular way. Patterns help us predict what comes next.

Colour the kites in a pattern of your choice.



When a pattern repeats again and again, it forms a loop.

We learned all of this in the previous class. Now, we will explore some new topics.

Let's Get Better

When solving puzzles, sometimes things don't work out right away. How can you stay calm and try again if you don't find the answer on the first try?

DECISION MAKING

Decision making is choosing what to do from different options. For example, when you decide what to eat for lunch or which game to play, you are making a decision.

INFO Byte

Your brain is like a computer. Every time you solve a puzzle, figure out a pattern or make a decision, you are teaching your brain to think faster and better.

Activity Corner

1. What to Wear? (**Instructions :** Imagine it is a rainy day. Which outfit would you choose to stay dry and warm?)

(a) (b) (c)

2. Healthy Snacks (**Instructions :** You are feeling hungry. Which snack is the healthiest choice?)

(a) (b) (c)

3. Choosing a Toy (**Instructions :** You have time to play with one toy. Which toy will you choose?)

(a) (b) (c)

4. Crossing the Street (**Instructions :** You want to cross the street. What should you do first?)

(a) Run across quickly
(b) Look both ways and wait for the signal
(c) Ask a friend to cross first

DECODING

Decoding is figuring out a secret message to understand its real meaning. It is like solving a puzzle where you change symbols, numbers, or letters into words you can read. Let us understand it with the help of an example :

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

Look at the table given above. We have assigned numbers to each alphabet. Now we will use numbers instead of alphabets for making a word. Like, we will use 1 in place of A, 2 in place of B and so on. For example, take the word APPLE :

So, 1, 16, 16, 12, 5 will denote the word APPLE.

A	P	P	L	E
↓	↓	↓	↓	↓
1	16	16	12	5

Decode the following :

C	L	A	S	S

T	I	G	E	R

4	1	14	3	5

16	1	20	20	5	18	14

So, whenever we figure out the real meaning of a secret message, it is called decoding.

Exercise

I. Tick (✓) the correct option :

1. What is decision making?
   (a) Choosing a game to play
   (b) Making choices by identifying information
   (c) Drawing pictures
   (d) All of the above
2. What does 'Yes' mean in decision making?
   (a) Do something if a condition is false
   (b) Do something if a condition is true
   (c) Always say yes
   (d) None of the above
3. Which sequence is pattern?
   (a) Red, blue, yellow, red, blue, yellow
   (b) Red, yellow blue, green
   (c) Blue, yellow, green, purple
   (d) Green, yellow, purple, red
4. What is a loop in computer language?
   (a) A repeating action
   (b) A type of fruit
   (c) A computer game
   (d) A drawing tool
5. What does 'decoding' mean?
   (a) Painting a picture
   (b) Drawing a pattern
   (c) Changing a secret message to its original meaning
   (d) Playing games

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A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

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A	P	P	L	E
↓	↓	↓	↓	↓
1	16	16	12	5

Decode the following :

C	L	A	S	S

T	I	G	E	R

4	1	14	3	5

16	1	20	20	5	18	14

So, whenever we figure out the real meaning of a secret message, it is called decoding.

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   (c) Blue, yellow, green, purple
   (d) Green, yellow, purple, red
4. What is a loop in computer language?
   (a) A repeating action
   (b) A type of fruit
   (c) A computer game
   (d) A drawing tool
5. What does 'decoding' mean?
   (a) Painting a picture
   (b) Drawing a pattern
   (c) Changing a secret message to its original meaning
   (d) Playing games

II. 1. Write the correct sequence for growing a tree :

2. Write in correct sequence in descending order :

5 8 2 9 ⬡ > ⬡ > ⬡ > ⬡

III. Complete the following patterns :

1. 3, 6, 9, ..., ..., ...
2. ....
3. ....
4. ....

IV. Match the following :

Term	Explanations
1. Decision Making	(a) A repeating design or order (like red, blue, yellow in a sequence)
2. Pattern	(b) Making choices by identifying information
3. Loop	(c) Process of converting a secret message to its original meaning
4. Decoding	(d) Telling a computer to do something repeatedly
5. Sequence	(e) Order in which things are arranged or occur

Smart Thinking

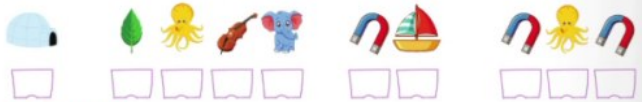
* **Read the instruction and solve the puzzle :**

10 less than 10	5 more than 10	10 more than 10
2 less than 10	10	2 more than 10
8 less than 10	5 less than 10	8 less than 10

* **Decode the hidden message below by writing the letter of each picture.**

A	B	C	D	E
F	G	H	I	J

Video frame 35 — 01:08



FOR THE TEACHER

Encourage students to practice finding patterns and solving problems through fun activities. Remind them that it's okay to make mistakes and to keep trying, as each step helps them learn and get better at thinking.

Instructions and Direction Sense

LET'S RECALL

Fill in the blanks :

Instructions to follow to cross a red light are :

1. We ..... when the light is red.
2. We ..... until the light turns green before crossing.
3. We ..... left and right.
4. We ..... when the light is green.

IN THIS CHAPTER

- What is Instruction?
- Direction Sense

In this chapter, we will learn how computers understand and follow instructions. Just like you follow directions to get to a place, computers follow instructions to perform tasks. We will learn how to give simple instructions to a computer and understand basic concepts.

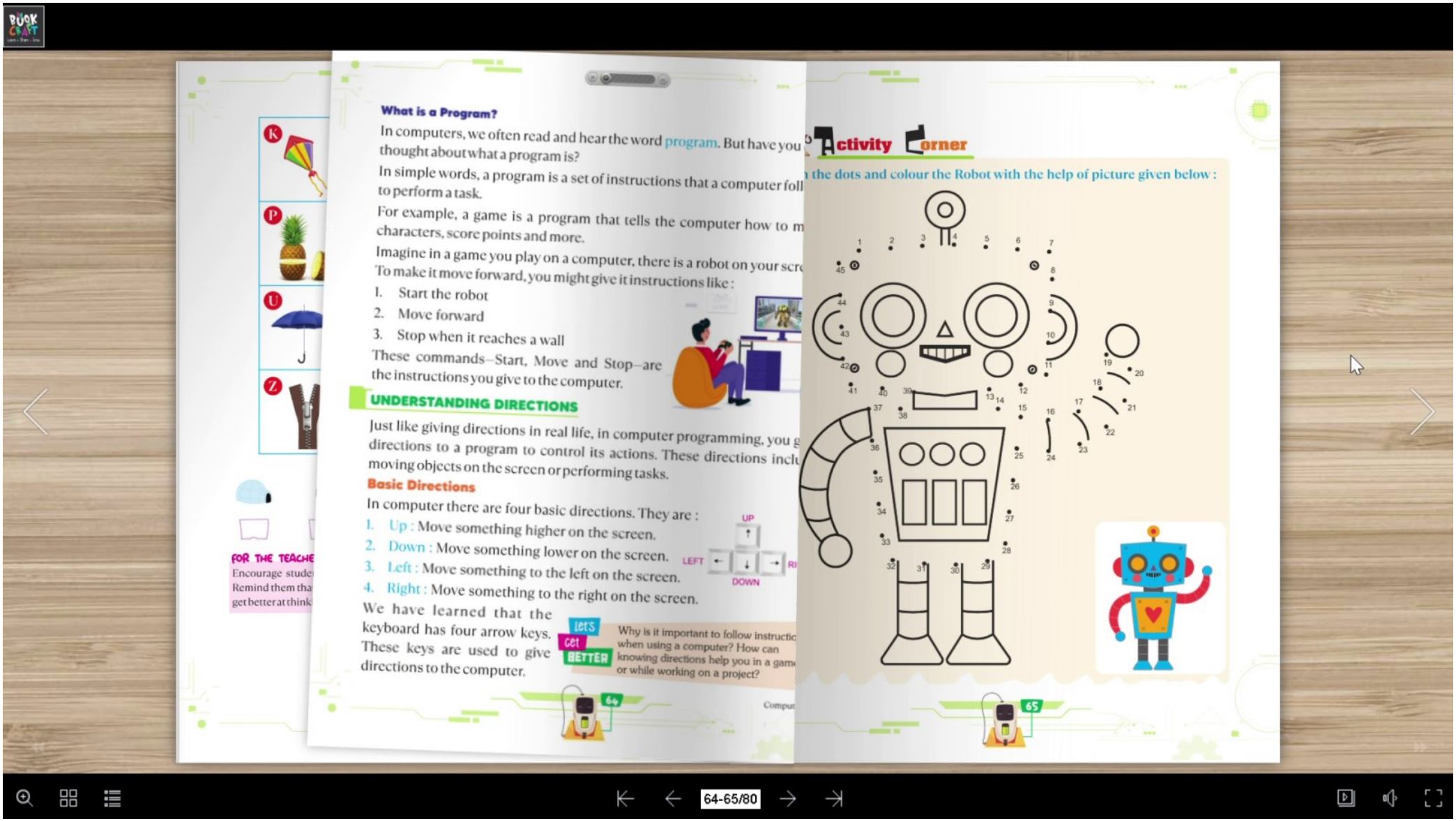
WHAT ARE INSTRUCTIONS IN COMPUTERS?

Instructions in computers are like steps in a recipe. They tell the computer what to do, step by step. When you use a computer program or game, you are giving it instructions on what to do next.

Opening a Program

When we want to open a program on the computer, we double-click on the program's icon. This action of double-clicking is the instruction we give to the computer.

Video frame 36 — 01:10



What is a Program?

In computers, we often read and hear the word program. But have you ever thought about what a program is?

In simple words, a program is a set of instructions that a computer follows to perform a task.

For example, a game is a program that tells the computer how to move characters, score points and more.

Imagine in a game you play on a computer, there is a robot on your screen. To make it move forward, you might give it instructions like :

1. Start the robot
2. Move forward
3. Stop when it reaches a wall

These commands–Start, Move and Stop–are the instructions you give to the computer.

UNDERSTANDING DIRECTIONS

Just like giving directions in real life, in computer programming, you give directions to a program to control its actions. These directions include moving objects on the screen or performing tasks.

Basic Directions

In computer there are four basic directions. They are :

1. Up : Move something higher on the screen.
2. Down : Move something lower on the screen.
3. Left : Move something to the left on the screen.
4. Right : Move something to the right on the screen.

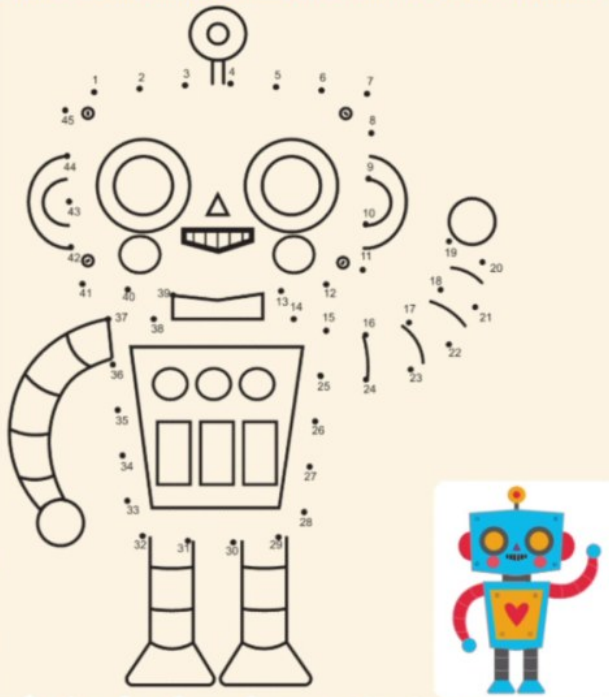
We have learned that the keyboard has four arrow keys. These keys are used to give directions to the computer.

Let's Get BETTER

Why is it important to follow instructions when using a computer? How can knowing directions help you in a game or while working on a project?

Activity Corner

Join the dots and colour the Robot with the help of picture given below :



Exercise

I. Tick (✓) the correct option :

1. What is an instruction in a computer?
   (a) A game character
   (b) A set of steps for the computer to follow
   (c) A computer part
   (d) A type of computer virus
2. What do you do to open a program on the computer?
   (a) Click once on the icon
   (b) Double-click on the icon
   (c) Press the spacebar
   (d) Use the mouse scroll wheel
3. Which of the following is NOT a basic direction in computer programming?
   (a) Up (b) Down
   (c) Forward (d) Left
4. What does the 'Stop' command do in a computer program?
   (a) It makes the computer turn off
   (b) It tells the computer to pause or end an action
   (c) It opens a new program
   (d) It moves the character faster

II. Write 'T' for true and 'F' for false :

1. Instructions in computers are similar to steps in a recipe.
2. To open a program, you need to single-click on the program's icon.

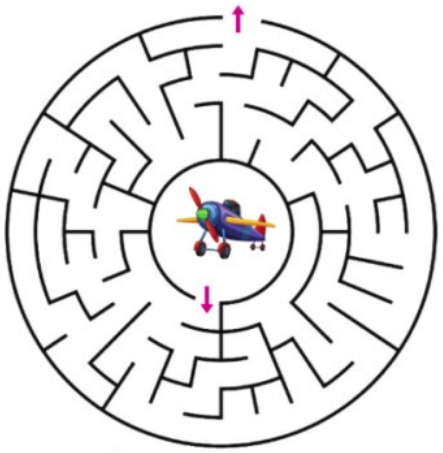
3. The four basic directions in computer programming are Up, Down, Left and Right.
4. A program is a random set of commands that does not have a specific purpose.

III. Answer the following questions :

1. What is a program in the context of computers?
2. Name the four basic directions used in computer programming.
3. How can you make a robot move forward in a game?

Smart Thinking

* **What path will you choose for the helicopter to fly through the maze and reach the exit?**



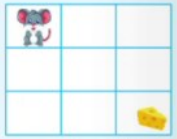
Lab Activity

A. Simple Drawing Program

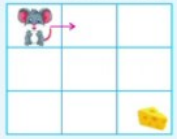
1. Open a simple drawing program like Tux Paint on the computer.
2. Use directions to move the cursor : Click and drag the mouse to draw shapes (up, down, left, right).

B. Create a Sequence

Mickey wants to eat the cheese. Let us create a sequence of instructions to help Mickey to reach it.



Step 1 : Move one step right.



Step 2 : Move one step right.

Step 3 : Move one step down.

Step 4 : Move one step down.

Note : Mickey can move only one box at a time.

Now create some other sequences yourselves to help mickey to reach the cheese.

Just like you follow directions to get somewhere, computers follow instructions to complete tasks. By understanding how to give and follow these instructions, you can start creating your own programs and games!

9 CHAPTER

AI and Robotics

Let's Recall

Which of the following is an intelligent machine?

In this Chapter

- Artificial Intelligence
- Robots
- AI in Everyday Life : Voice Assistants, Chat bots
- Uses of Robots

Let's Start with a Question :

If Nitin has 8 chocolates and Akash has 5 chocolates, how many chocolates do they have in total?

When you read the question, you probably started thinking and figured out the answer using what you already know. This thinking process is called Natural Intelligence or Human Intelligence – our ability to think, learn, and solve problems.

Now, imagine a machine that can think and do tasks like a person. This is called Artificial Intelligence (AI). A machine with AI is known as a smart machine because it can perform tasks like a human, making life easier and more enjoyable.

Lab Activity

A. Simple Drawing Program

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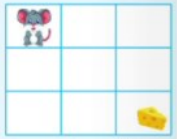
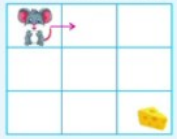


Step 1 : Move one step right.



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Step 3 : Move one step down.

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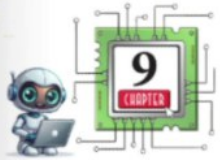

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AI in Everyday Life

AI is all around us, helping in ways we may not even notice. Let's look at some examples of how AI is part of our everyday life.

Voice Assistants

Voice assistants are apps that understand and respond to your voice. Known as AI or digital assistants, they help you with tasks using voice commands. Many smartphones have these features built-in. Let's explore some popular voice assistants:

OK Google

OK Google, made by Google, is a voice assistant that helps you manage tasks on the go, like setting reminders and controlling smart devices. Just say "OK Google" and ask for what you need.

Siri

Siri is Apple's voice assistant for iPhone users. It helps you do daily tasks just by listening to your commands. Talking to Siri can feel like talking to a helpful friend.

Amazon Alexa

Alexa is Amazon's voice assistant, available on many of their devices. You can ask it questions, play music, or control smart home devices like lights and thermostats by speaking. For example: Say, "What's the weather?" and Alexa will respond.

Biometric Technology

Biometric technology uses unique traits, like your face or fingerprint, to keep devices secure. It ensures that only you can unlock your device, keeping it safe.

INFO Byte

John McCarthy is considered as the father of Artificial Intelligence.

Face Recognition

Face recognition is a common security feature on phones. Your device remembers your face pattern and checks it when you try to unlock it, making sure only you can access it.

WHAT IS A ROBOT?

A robot is a machine that can move and do tasks on its own or with some help from a human. They follow step-by-step instructions, just like a recipe or a set of rules.

Robots can look like humans, animals, or just be simple machines. They can do many different jobs, like cleaning, building things, or even playing games. Robots use sensors and computers to understand what they need to do and how to do it.

Robots	Human
Robots are machines created by people.	Humans are natural and living beings.
Robots do not have emotions or feelings.	Humans have emotions, like happiness, sadness, and love.
Robots follow instructions given by humans.	Humans can think, learn from experiences and make decisions on their own.
Robots do not get tired and can work continuously.	Humans need rest and sleep to recharge.
Robots can not learn from personal experiences, but they can be programmed to improve based on data.	Each human has unique experiences and knowledge.

Activity Corner

Name the following AI enabled devices / apps :

1 2 3

....

Types of Robots

Industrial Robots : These robots work in factories. They help make cars, assemble products, and do heavy lifting.

In Factory

To Hospital

Medical Robots : These robots help doctors and patients. They perform surgeries, deliver medicines, and assist in rehabilitation.

Entertainment Robots : These robots provide fun and amusement. They can play games, act as toys, or perform in shows.

To Play

To Security

Military Robots : These robots assist in defense and security. They can defuse bombs, carry supplies, and conduct surveillance.

Let's Get Better

How do you think artificial intelligence, like voice assistants and robots, can make our daily lives easier? Can you think of a time when you used a robot or voice assistant to help you?

Exercise

I. Tick (✓) the correct option :

1. What is a voice assistant?
   - (a) A type of robot ☐
   - (b) An app that understands and responds to voice commands ☐
   - (c) A computer game ☐
   - (d) A type of smartphone ☐
2. Which of these is an example of a voice assistant?
   - (a) OK Google ☐ (b) YouTube ☐
   - (c) Microsoft Word ☐ (d) Instagram ☐
3. What does biometric technology use to identify people?
   - (a) Names ☐
   - (b) Personal characteristics like face and fingerprints ☐
   - (c) Birth dates ☐ (d) Voice commands ☐
4. Which type of robot is used in factories to help make cars?
   - (a) Medical robots ☐ (b) Service robots ☐
   - (c) Industrial robots ☐ (d) Entertainment robots ☐

II. Write 'T' for true and 'F' for false :

1. Robots can get tired and need rest. ☐
2. Siri is a voice assistant developed by Google. ☐
3. Biometric technology can use fingerprints to identify a person. ☐
4. Entertainment robots can perform in shows. ☐
5. Humans cannot learn from personal experiences. ☐

III. Fill in the blanks with the help of the given box :

Face
Instructions
Sensors
Medical
Amazon

1. Alexa is a voice assistant developed by ..... .
2. Robots follow ..... given by humans.
3. ..... robots help doctors and patients.
4. Biometric technology uses personal characteristics like ..... and fingerprints.
5. Robots use ..... and computers to understand tasks.

IV. Answer the following questions :

1. What is a robot?
2. How do voice assistants like Siri and Alexa help us in everyday life?
3. What is the main difference between human intelligence and artificial intelligence?
4. Name three types of robots and explain their uses.

Lab Activity

(a) Use a smartphone with a voice assistant (like Siri, Alexa, or OK Google). Ask it various questions or give it commands, and observe how it responds. Discuss how it helps in everyday tasks.

(b) Create a simple robot using a small kit or LEGO. Discuss how the robot follows instructions and what tasks it can perform.

FOR THE TEACHER

Show students AI-enabled gadgets to help them better understand the concepts.

Periodic Test Term-1

(Based on Chapters 1 to 3)

Time .....

Marks .....

Note : All questions are compulsory.

A. Tick (✓) the correct option :

1. Monitor looks like a ..... ?

(a) (b) (c) (d)

2. A computer stores a large amount of ..... .

(a) money (b) information

(c) water (d) None of these

3. Which of the following characters will display with this combination?

(a) 2 (b) @ (c) @2 (d) 2@

B. Fill in the blanks :

1. A computer is a ..... device.
2. We use a computer at home to play ..... .
3. To type capital letter G : ..... + G

C. Write 'T' for true and 'F' for false.

1. UPS stands for Uninterruptible Power Support.
2. Banks use computers to manage accounts and details of customer.
3. The keys that show the letters A to Z are called number keys.

D. Answer the following questions :

1. What do you mean by computer?
2. Write any two uses of computer.
3. How many number keys are present on the keyboard?

Half Yearly Test Paper

(Based on Chapters 1 to 5)

Time .....

Marks .....

Note : *All questions are compulsory.*

A. Tick (✓) the correct option :

1. Which key is used to move the cursor?
   (a) Delete ☐ (b) Arrow ☐
   (c) Backspace ☐ (d) Shift ☐
2. Both CDs and DVDs are ..... discs and are the same size.
   (a) Round ☐ (b) Square ☐ (c) Rectangle ☐ (d) Triangle ☐
3. Which of the following is a printer?
   (a) ☐ (b) ☐
   (c) ☐ (d) ☐
4. Which of the following things cannot be done with the help of a computer?
   (a) Play games ☐ (b) Share feelings ☐
   (c) Solve sums ☐ (d) Shopping ☐

B. Fill in the blanks :

1. ..... devices allow us to enter data and information in the computer.
2. ..... is known as the brain of computer.
3. ..... get tired after working sometime.
4. In shops, computers are used for making ..... .

C. What is the child doing in given pictures?

....

D. Match the following symbols with their names.

	Symbol	Name
1.	?	semicolon
2.	;	parentheses
3.	!	question Mark
4.	( )	comma
5.	,	exclamation mark

E. Who am I?

1. I have many buttons, called keys. .....
2. I help you record sound. .....
3. I help scan pictures. .....
4. I help you listen to music on the computer. .....
5. I help you store data and information.
   I am fixed inside the CPU cabinet. .....

F. Answer the following questions :

1. What is the full form of IPO?
2. Write the names of any two input and output devices.
3. Write two difference between human and computer.
4. Write any two disadvantages of a computer.

Periodic Test Term-2

(Based on Chapters 6 and 7)

Time .....

Marks .....

Note : *All questions are compulsory.*

A. Tick (✓) the correct option :

1. The largest part of the Tux Paint window is ..... .
   (a) drawing canvas (b) toolbar
   (c) colours (d) None of these
2. What is decision making?
   (a) Choosing a game to play
   (b) Making choices by identifying information
   (c) Drawing pictures (d) All of the above
3. ..... tool is used to type text on the drawing canvas.
   (a) Save (b) Brush (c) Paint (d) Abc Text

B. Fill in the boxes :

1. ..... tool is used to draw free hand drawings.
2. 3, 6, 9, ... , ... , ...
3. ..... is used to paste pre-drawn images in the picture.

C. Write 'T' for True and 'F' for False :

1. The keyboard has number 1 to 100 on all the keys.
2. Decoding is figuring out a secret message to understand its real meaning.
3. Decision making is choosing what to do from different options.

D. Answer the following questions :

1. Write the steps to open Tux Paint.
2. What is a loop in computer language?
3. What does 'decoding' mean?

Annual Test Paper

Time .....

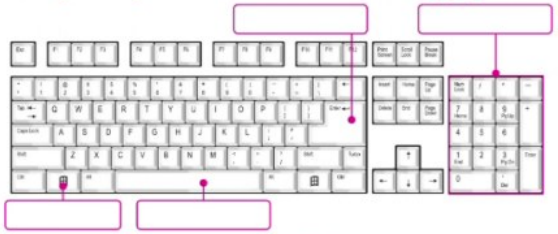
Marks .....

Note : *All questions are compulsory.*

A. Tick (✓) the correct option :

1. What is an instruction in a computer?
   (a) A game character
   (b) A set of steps for the computer to follow
   (c) A computer part (d) A type of computer virus
2. What is a voice assistant?
   (a) A type of robot
   (b) An app that understands and responds to voice command
   (c) A computer game (d) A type of smartphone
3. What do you do to open a program on the computer?
   (a) Click once on the icon (b) Double-click on the icon
   (c) Press the spacebar (d) Use the mouse scroll wheel
4. Which type of robot is used in factories to help make cars?
   (a) Medical robots (b) Service robots
   (c) Industrial robots (d) Entertainment robots

B. Identify the keys and write their names in the appropriate box :



C. Fill in the blanks :

1. Click on ..... button, to save drawing.
2. ....
3. ..... in computers are like steps in a recipe.
4. Alexa is a voice assistant developed by ..... .

D. Write input, process and output under the pictures given below :

....

E. Unscramble the word :

1. O L O R U C .....
   The palette available at the bottom of the main screen.
2. C T O R S E L E .....
   It shows the different options when a tool is selected.
3. R A S E E R .....
   This tool is used to erase a part or whole of the drawing.

F. Answer the following questions :

1. What is a program in the context of computers?
2. Name the four basic directions used in computer programming.
3. How can you make a robot move forward in a game?

C. Fill in the blanks :

1. Click on ..... button, to save drawing.
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Mastering CODING

AI AND ROBOTICS

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